

Capstone requirements overview

This document summarizes functional requirements (what the system does), non-functional requirements (how well it must perform).

Quick guide: functional vs non-functional

Functional requirements describe behavior and features — for example, saving an exam configuration, submitting an answer, or returning a structured grade.

Non-functional requirements describe qualities and constraints — for example, page load time, uptime targets, accessibility, logging policy, and operator documentation.

Technical constraints

- Implementation stack: Python backend, web UI, HTTP API, database (e.g. SQLite or Postgres).
- LLM: primary provider Together (or equivalent); subject to network, quotas, latency, and cost.
- Architecture: English prompt templates in code → filled at runtime → model may return Python source → execute only in a sandbox or heavily restricted environment → validate against versioned schemas before use.
- Security: model output is untrusted; no full-privilege exec on the host; limits on CPU, memory, and time.
- LLM calls are stateless: each request must include the context the model needs (question, rubric, follow-up history, etc.).
- Testing: automated tests often use a mock LLM; live integration tests need keys and tolerate provider variability (retries, p95).
- Compliance: FERPA/PII handling, retention, and export may be constrained by institutional policy (see SRS open items).

Functional requirements

Product goal

G-1 — Question variation: with `vary_questions` enabled, the system targets diverse questions (e.g. not byte-identical to another student's in the same configuration within seven days; acceptance is partly stochastic).

Professor configuration

FR-PROF-1 — Create and save exam configuration: title, N questions (1–20), optional time limits, domain briefing, optional reference files, rubric style, weights summing to 1.

FR-PROF-2 — At least three independent constraint fields per question slot, passed into generation and audit-logged.

FR-PROF-3 — Per-student time multipliers (1.0×–3.0×) for question and whole-exam limits.

FR-PROF-4 — `vary_questions` flag; prompts discourage repeating recent instances for the same config.

LLM — question generation

FR-GEN-1 — Model returns Python assigning a named result; execute under security rules; validate to `exam_gen_v1`; invalid responses trigger retries.

FR-GEN-2 — Generated content includes background, essay question, rubric (≥ 3 criteria with points), expected concepts, metadata.

FR-GEN-3 — Retries with exponential backoff (up to three) on failure.

FR-GEN-4 — Generation latency budget ($p_{95} \leq 45$ s under nominal staging conditions).

Student examination

FR-STU-1 — Session bound to one student and one exam config; no answers after session is closed, expired, or finally submitted.

FR-STU-2 — Question order is fixed or random, chosen at session start and stable across refresh.

FR-STU-3 — Essay answers up to 20,000 UTF-8 characters; enforce on client and server.

FR-STU-4 — Autosave drafts on a defined schedule.

FR-STU-5 — Optional per-question and whole-exam timers; server-authoritative expiry (± 5 s).

Grading

FR-GRADE-1 — Grading prompt includes question, rubric, background, answer, optional timing, weighting; auditable.

FR-GRADE-2 — Grading uses same Python-exec pattern; validate `exam_grade_v1` (scores, total 0–100, satisfactory vs threshold, explanation, confidence).

FR-GRADE-3 — After failures, mark `needs_human_review`; no numeric final until professor resolves.

Follow-up rounds (optional path)

FR-FU-1 — If not satisfactory and follow-ups enabled, request follow-up question and rubric within limits.

FR-FU-2 — Final grade considers all student turns; one terminal `exam_grade_v1`.

FR-FU-3 — Persist State P: grades, rubrics, backgrounds, prompt metadata, student text, timestamps.

Persistence

FR-DB-1 — Store finalized questions under (`student_id`, `exam_session_id`, `question_index`); index by session.

FR-DB-2 — Configurable retention and purge of old records.

Final exam grade

FR-FINAL-1 — When all questions resolved, build summaries for final grade computation.

FR-FINAL-2 — LLM returns Python $\rightarrow$ `exam_final_v1` (final percentage, optional letter grade, explanation).

FR-FINAL-3 — At least two professor-selectable grading schemes (e.g. pure weighted average vs LLM-adjusted composite).

Results and disputes

FR-DISP-1 — After finalization, show student final percentage, per-question scores, truncated explanations.

FR-DISP-2 — Disputes only after a configurable delay; caps on length and count per question.

FR-DISP-3 — LLM dispute triage schema; no automatic grade change unless professor enables it.

Professor review

FR-REVIEW-1 — Browse/filter sessions with stated performance targets on reference data.

FR-REVIEW-2 — One consolidated view of prompts (secrets redacted), rubrics, backgrounds, answers, model output, timings, aggregates.

FR-REVIEW-3 — Export session (JSON required in v1; PDF optional); JSON matches `exam_export_v1`.

Security (behavioral)

FR-SEC-1 — Never run LLM-generated Python with full host privileges; subprocess sandbox or restricted exec with documented limits.

FR-SEC-2 — Authenticate professor routes; student routes use session tokens bound to student and config with bounded lifetime.

Optional / advanced (if implemented)

ADV-CURVE-1 — Cohort curve with minimum sample size and clear fallback when too few responses.

ADV-DISC-1 — Discussion-based testing with transcript limits and token budget per grading call.

Non-functional requirements

NFR-PERF-1 — Student UI: Largest Contentful Paint ≤ 2.5 s under a defined throttled network profile (median of measured runs).

NFR-REL-1 — Core API availability $\geq 99.5\%$ over a two-week pilot (synthetic checks), excluding provider outages per policy.

NFR-AUDIT-1 — Immutable audit events for key actions; correlation IDs; very high durability on committed transactions.

NFR-LOG-1 — No raw API secrets in logs; logging full student answers off by default (opt-in).

NFR-ACC-1 — WCAG 2.1 Level AA on primary student flows; automated axe scan with zero critical violations on those flows.

NFR-I18N-1 — Externalized user-visible strings; English (en-US) for version 1.

NFR-DOC-1 — Operator documentation: deploy, environment variables, backup/restore, incident runbook; page limit per SRS.